

ENGN1560 Transmission Lines / Low Pass Filter Lab #2

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10/09/2026

Introduction

In this lab, we took the 50 Ω microstrip line designed in Lab 1 and used it to test three common microwave techniques in Ansys HFSS. In part 2.1, the line was matched to a complex load of 80 - j60 Ω at 1 GHz using a single open stub. In part 2.2, a 40 Ω load was matched to a 75 Ω line at 2 GHz using a quarter-wave transformer. In part 2.3, I designed and simulated a sixth-order stepped-impedance low-pass filter based on Pozar Example 8.6, with a target cutoff of 2.5 GHz on FR4. The filter was then built and measured on the VNA during the in-person session.

Most of the results below are given in terms of VSWR (voltage standing wave ratio). A VSWR of 1 means all of the power sent down the line reaches the load. Higher values mean some of the power is being reflected back toward the source instead. As a reference point, a VSWR of 2 means about 11% of the power is reflected, and a VSWR of 3 means about 25% is reflected. Reflected power is wasted, and in a real system, it can also damage the transmitter. For the filter, results are given as S-parameters. S21 is how much a signal makes it through the filter, and S11 is how much is reflected back at the input.

Part 2.1: Single Stub Tuning

In this part, I reused the 50 Ω line from Lab 1 and changed the port 2 impedance to 80 - j60 Ω . Before running the simulation, I calculated the expected mismatch from the load reflection coefficient:

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{(80 - j60) - 50}{(80 - j60) + 50} = \frac{30 - j60}{130 - j60} = 0.469 \angle -38.7^\circ$$

$$\text{VSWR} = \frac{1.469}{0.531} = 2.76$$

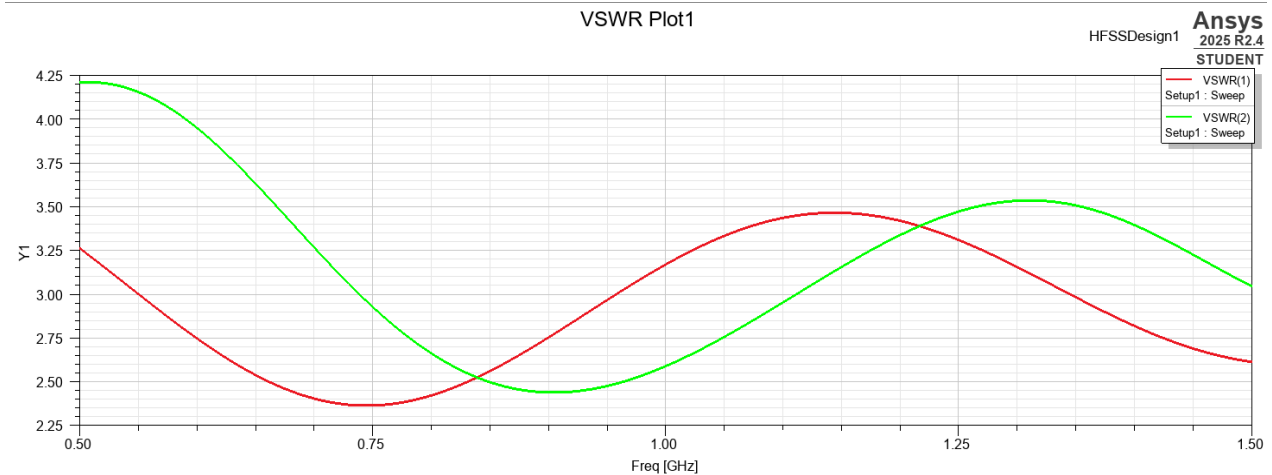


Figure 1: *VSWR at port 1 (red) and port 2 (green) of the original 50 Ω line with port 2 set to 80 - $j60 \Omega$, 0.5 to 1.5 GHz.*

Figure 1 shows the VSWR rising and falling between about 2.4 and 3.5 across the frequency range. At 1 GHz it is about 3.2, which is in the same range as the 2.76 I calculated by hand. The VSWR moves up and down with frequency because the simulated line is not exactly 50 Ω . Small reflections at both ends of the line add together or cancel depending on the length of the line in wavelengths, and that changes with frequency. A VSWR of 3.2 means about 27% of the power going toward the load is reflected back, which is a significant loss.

Stub Design

The stub design came from my SimSmith solution for homework problem 1.6. Dividing the load by 50 Ω gives a normalized load of $Z_L = 1.6 - j1.2$. For a series stub, the stub has to be placed at a distance d from the load where the real part of the normalized impedance looking toward the load is equal to 1. The closest solution to the load is $d = 0.0325\lambda$ and $l = 0.380\lambda$ with an open circuited stub.

There is a second solution at $d = 0.360\lambda$ with a 0.120λ stub, but I used the first one because the lab asks for the stub to be as close to the load as possible. To convert to physical lengths, I used the wavelength on the 50 Ω FR4 line at 1 GHz. The line has $\epsilon_r = 4.4$, $h = 1.6\text{mm}$, width of about 2.96 mm, and an effective permittivity of about 3.32. Thus, the calculated length of the open stub is 62.6 mm.

I drew the stub as a separate box with the same width as the main line and joined the two with the Unite command. I then adjusted the stub length in HFSS to move the lowest VSWR point onto 1 GHz. This tuning was needed because the junction where the stub meets the line, and the

open end of the stub, both add a small amount of extra capacitance that SimSmith does not account for.

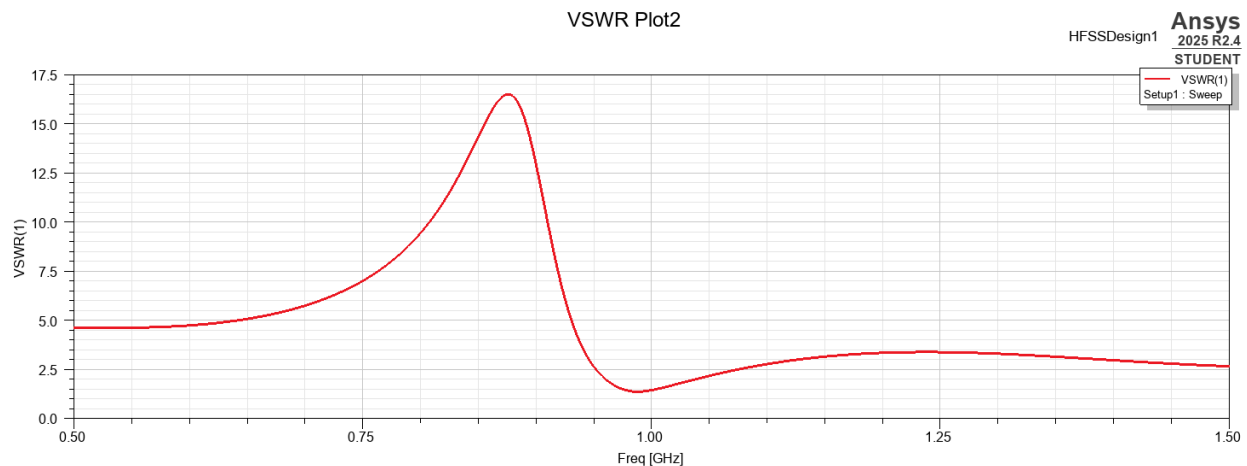


Figure 2: *VSWR at port 1 after adding the tuned open stub, 0.5 to 1.5 GHz.*

Figure 2 shows the result after tuning. The VSWR now drops to about 1.35 at roughly 0.99 GHz. This corresponds to a reflection coefficient of about 0.15, so only about 2% of the power is reflected. Compared to the unmatched line in Figure 1, the stub reduced the reflected power at 1 GHz by more than a factor of ten.

The match only works over a narrow range of frequencies. The VSWR stays below 2 only from about 0.96 to 1.04 GHz. Outside that range it climbs back up to between 2.5 and 4.6. There is also a sharp spike of about 16.5 near 0.875 GHz, where the stub's reactance changes quickly with frequency and creates a large mismatch. The stub cancels the load's reactance at one frequency only, so it works well for a single-frequency signal but would not work for a signal that covers a wide band. The remaining VSWR of 1.35 instead of 1.0 is likely due to effects mentioned above. Adjusting the stub position even further along with its length could potentially bring it closer to 1.

Part 2.2: Quarter-Wave Transformer

Unmatched Line

To set up the problem 1.7 scenario, I set port 1 to 75 Ω , representing the line, and port 2 to 40 Ω , representing the load. The original 50 Ω microstrip was left in between. The direct mismatch between 75 Ω and 40 Ω gives:

$$\text{VSWR} = 75/40 = 1.875$$

With a $50\ \Omega$ line in between, the VSWR should move between two limits as the frequency changes. When the line is a whole number of half wavelengths long, port 1 sees the $40\ \Omega$ load directly and the VSWR is 1.875. When it is an odd number of quarter wavelengths long, the line itself transforms the load to $62.5\ \Omega$. That gives a VSWR of $75/62.5 = 1.2$.

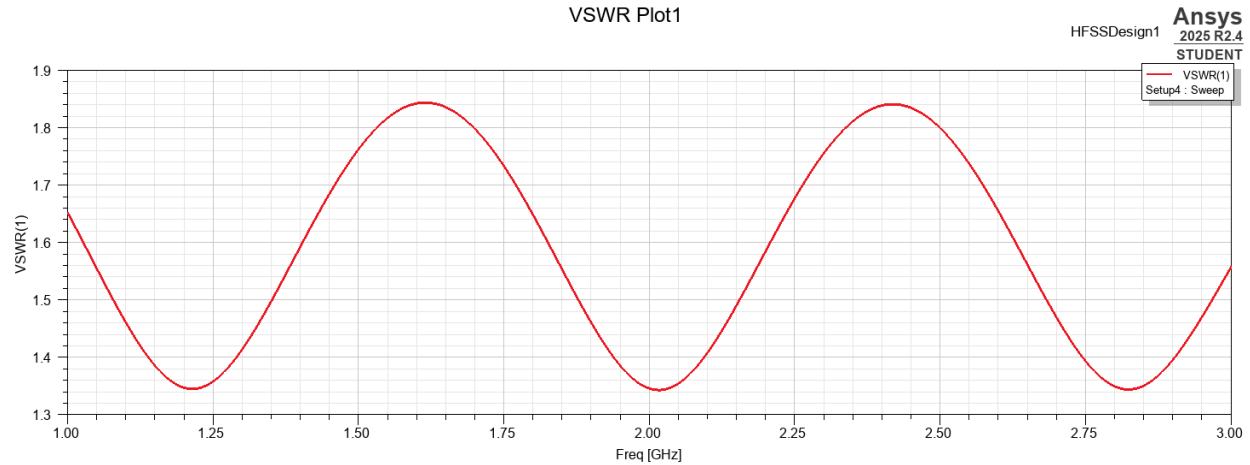


Figure 3: VSWR at port 1 of the original $50\ \Omega$ line with port 1 set to $75\ \Omega$ and port 2 set to $40\ \Omega$, 1 to 3 GHz.

Figure 3 shows this pattern. The VSWR peaks at about 1.84, very close to the expected 1.875. It dips to about 1.34 at 1.22, 2.02, and 2.83 GHz. The dips are a bit higher than the ideal 1.2, which makes sense because the simulated line is not exactly $50\ \Omega$ and has some loss. At 2 GHz, the VSWR is about 1.34, which means about 2% of the power is reflected.

Transformer Design

A quarter wave transformer matches two real impedances by inserting a section that is a quarter wavelength long. The impedance of that section is given by:

$$Z_1 = \sqrt{Z_0 Z_L} = \sqrt{(75)(40)} = 54.77\ \Omega$$

I found the width and length using a transmission line calculator (see Figure 4) with the same FR4 parameters as Lab 1 at 2 GHz. This gave a width of about 2.52 mm and an effective permittivity of about 3.28, so the length of the section is ~ 20.7 mm.

$$\lambda_g/4 = \frac{c}{4f\sqrt{\epsilon_{eff}}} = \frac{3 \times 10^8}{4(2 \times 10^9)\sqrt{3.28}} = 20.7\text{ mm}$$

ϵ_r :	4.4	Tan δ :	0.02	GHz	Rho:	1		
Height:	63	mils	Thickness:	3.94	mils	Frequency:	2	GHz

Analysis/Synthesis Values

Width:	98.96542	mils	Zo:	54.77	Ohms
Length:	818.24377	mils	Angle:	90	degrees

ϵ_{eff} :		Loss:	0.129	dB
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Figure 4: Transmission line calculator results.

I placed the quarter-wave section at the port 2 end of the line, next to the 40 Ω load, and re-ran the simulation.

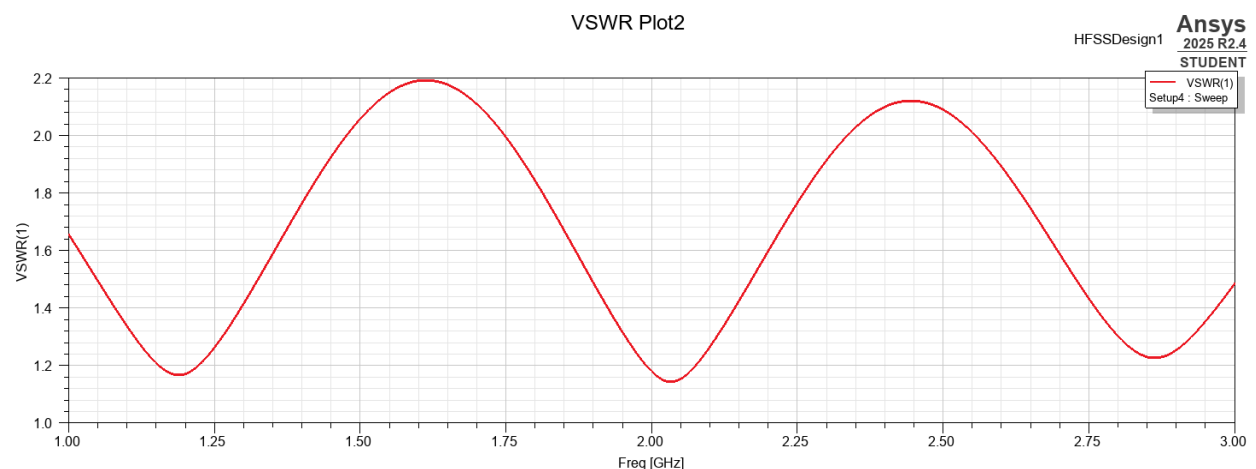


Figure 5: *VSWR at port 1 after adding the 54.77 Ω quarter-wave transformer; 1 to 3 GHz.*

Figure 5 shows a lower minimum of about 1.14 at 2.03 GHz, and about 1.17 at exactly 2 GHz. This is a reflection coefficient of about 0.07, meaning only about 0.4% of the power is reflected, compared to about 2% before. The minimum lands slightly above 2 GHz. This is likely due to rounding in the calculated width and length, and to the small extra capacitance at the points where the line width changes.

As with the stub, the improvement only holds near the design frequency. Away from 2 GHz, the peaks got larger, reaching 2.2 at 1.61 GHz and 2.1 at 2.45 GHz. This is because the transformer is only exactly a quarter wavelength long at 2 GHz. The rest of the line is still 50 Ω instead of 75 Ω , so there are now multiple impedance changes along the path. At some frequencies, the reflections from these changes add together instead of cancelling.

Part 2.3: Stepped-Impedance Low-Pass Filter

This filter follows Pozar Example 8.6. It is a sixth-order maximally flat low-pass filter with 50 Ω ports, a lowest practical line impedance of 20 Ω , and a highest of 120 Ω . In this type of filter, short sections of wide, low-impedance line act like capacitors connected to ground. Short sections of narrow, high-impedance line act like inductors in series. Alternating the two recreates a standard LC ladder low-pass filter using only copper traces.

I took the impedance and electrical length of each section directly from Pozar's solution. I then converted them to physical dimensions on our FR4 boards with $\epsilon_r = 4.4$, $\tan\delta = 0.02$, a substrate height of 63 mils (1.6 mm), a trace thickness of 3.94 mils (0.1 mm), and a frequency of 2.5 GHz.

I found the trace width for each impedance by using Pozar's microstrip design equations:

$$\frac{W}{d} = \begin{cases} \frac{8e^A}{e^{2A} - 2} & \text{for } W/d < 2 \\ \frac{2}{\pi} \left[B - 1 - \ln(2B - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left\{ \ln(B - 1) + 0.39 - \frac{0.61}{\epsilon_r} \right\} \right] & \text{for } W/d > 2, \end{cases}$$

where

$$A = \frac{Z_0}{60} \sqrt{\frac{\epsilon_r + 1}{2}} + \frac{\epsilon_r - 1}{\epsilon_r + 1} \left(0.23 + \frac{0.11}{\epsilon_r} \right)$$

$$B = \frac{377\pi}{2Z_0\sqrt{\epsilon_r}}.$$

For the 20 Ω sections, $A = 0.709$, and $B = 14.12$. This gives $W/d = 6.94$, so $W = 11.11\text{mm}$ (the $W/d > 2$ form applies). For the 120 Ω sections, $A = 3.447$, which gives $W/d = 0.255$, so $W = 0.41\text{mm}$ (the $W/d < 2$ form applies).

For length, I found the values by finding the effective dielectric constant given by:

$$\epsilon_e = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + 12d/W}}.$$

Which gives $\epsilon_e = 3.73$ for the 20 Ω lines, and 2.95 for the 120 Ω lines. The guided wavelength at the 2.5 GHz cutoff is $\lambda_g = \frac{c}{f\sqrt{\epsilon_e}}$ which is 62.14 mm for the 20 Ω lines and 69.92 mm for the 120 Ω lines. Each physical length is then $l = \frac{\beta l}{360^\circ} (\lambda_g)$. For section 1, the length was then calculated to be 2.04 mm. This calculation was repeated for all sections, with results detailed below.

Section	Impedance Ω	Bl (deg)	Width (mm)	Length (mm)
1	20	11.8	11.1	2.04
2	120	33.8	0.41	6.57
3	20	44.3	11.1	7.65
4	120	46.1	0.41	8.95
5	20	32.4	11.1	5.59
6	120	12.3	0.41	2.39

Figure 6: Stepped impedance filter section dimensions on FR4 at 2.5 GHz. The 50 Ω feed lines on each end are 3.05 mm wide. The equations used assume an infinitely thin conductor and ignore the small discontinuities where the trace width changes. Both are sources of error that I found when comparing the simulation and measurement.

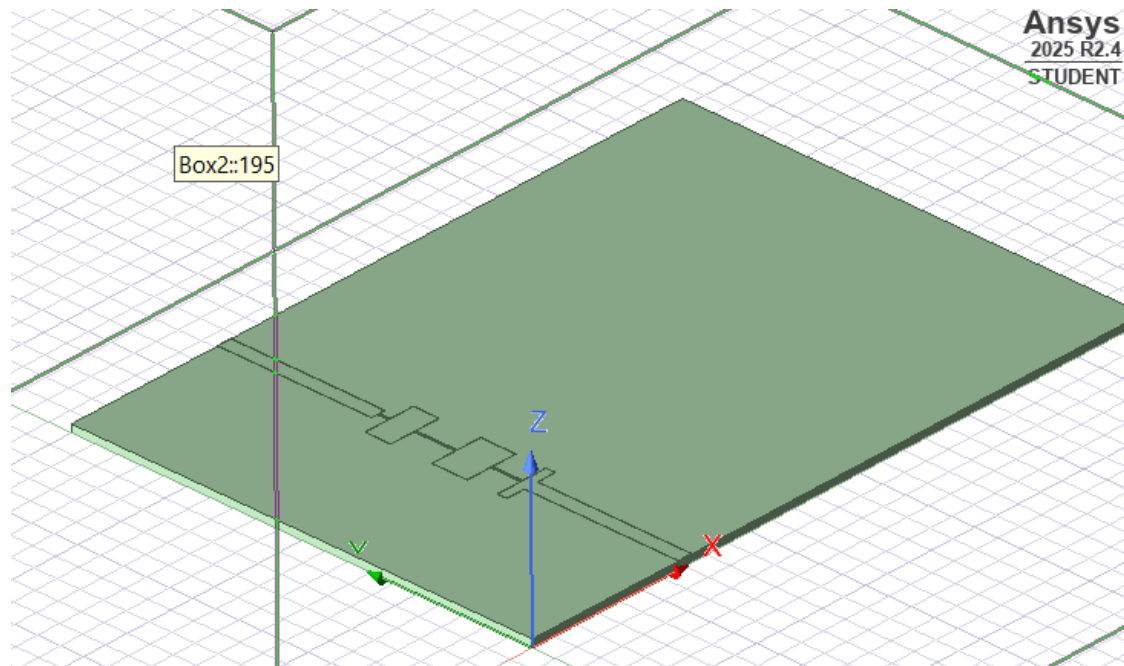


Figure 7: HFSS 3D model of the stepped-impedance low-pass filter on the FR4 substrate.

Figure 7 shows the finished model. The wide pads are the capacitor sections and the thin traces are the inductor sections. There is a $50\ \Omega$ feed line and port at each end.

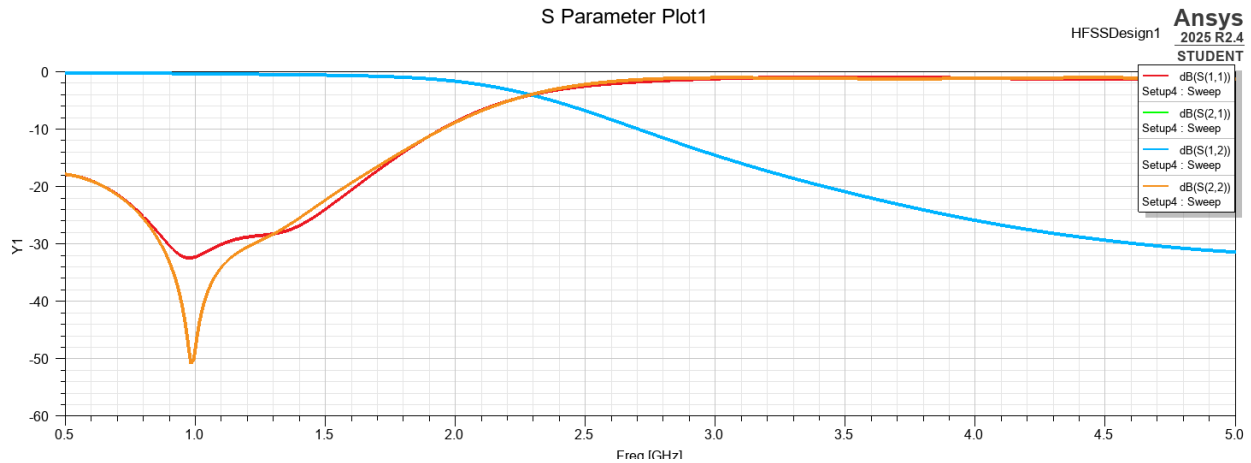


Figure 8: Simulated S-parameters of the low-pass filter from 0.5 to 5 GHz. S11 is red, S12 is blue, S22 is orange, and S21 (green) is hidden directly under S12.

The blue curve in Figure 8 shows how much signal passes through the filter. S21 is identical and is hidden underneath it, which is expected for a passive circuit like this one. Below about 2 GHz, the signal stays within about 1.5 dB of 0 dB, meaning nearly all of it gets through. The -3 dB point, where half the power gets through, is at about 2.2 GHz. Above that, the signal drops steadily.

Frequency (GHz)	Transmission (dB)
2.5	~ -7
3.0	~ -14.5
4.0	~ -26
5.0	~ -31.5

Figure 9: Simulated transmission above the cutoff.

The 26 dB of loss at 4 GHz meets the requirement from Pozar's example of more than 20 dB at 4 GHz. At that level, less than 0.3% of the power at 4 GHz reaches the output.

The red and orange curves show the reflections at each port. Up to about 1.5 GHz they stay below -20 dB, meaning less than 1% of the power is reflected. Both have deep dips near 1 GHz, about -32 dB for S11 and -51 dB for S22. Above the cutoff they rise toward 0 dB. This shows

that the filter blocks high frequencies mostly by reflecting them back toward the source, not by absorbing them. The small loss of a few tenths of a dB in the passband, even where the reflections are very low, comes from loss in the FR4 and the copper.

The cutoff ended up about 12% below the 2.5 GHz target. The most likely reason is that every sudden change in trace width adds a small amount of extra capacitance that the design equations do not include. Extra capacitance lowers the cutoff of a low-pass filter. The design equations also treat each short section as an ideal capacitor or inductor, which is only an approximation, especially for the 11 mm wide pads. Shortening the capacitor sections slightly would likely move the cutoff back up toward 2.5 GHz.

Part 2.3: In person build and VNA Measurement:

I built the filter on an FR4 board using copper foil tape cut to the dimensions in Figure 6, with an SMA connector at each end of the 50 Ω feed lines (Figure 10). Starting from the left port, the layout is 50 Ω feed, 11.1 x 2.04 mm capacitor pad, 0.41 mm x 6.57 inductor trace, 0.11 x 7.65 mm capacitor pad, 0.41 mm x 8.95mm inductor trace, 11.1 x 5.59 mm capacitor pad, 0.41 mm x 2.39 mm inductor trace, 50 Ω feed.

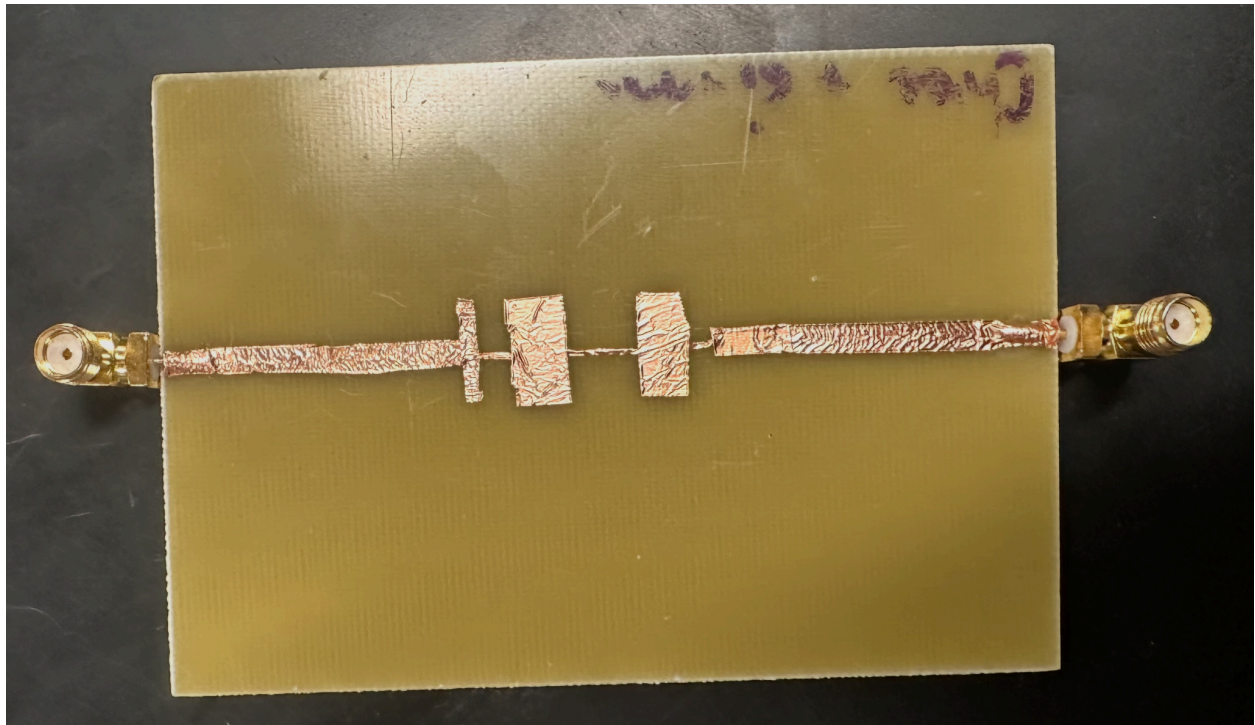


Figure 10: Completed stepped-impedance low-pass filter on FR4.

We then measured S21 with the VNA from 100 kHz to 5 GHz. Figure 11 shows the completed sweep. Market 1 sits at 3.0025 GHz and reads -36.63 dB.

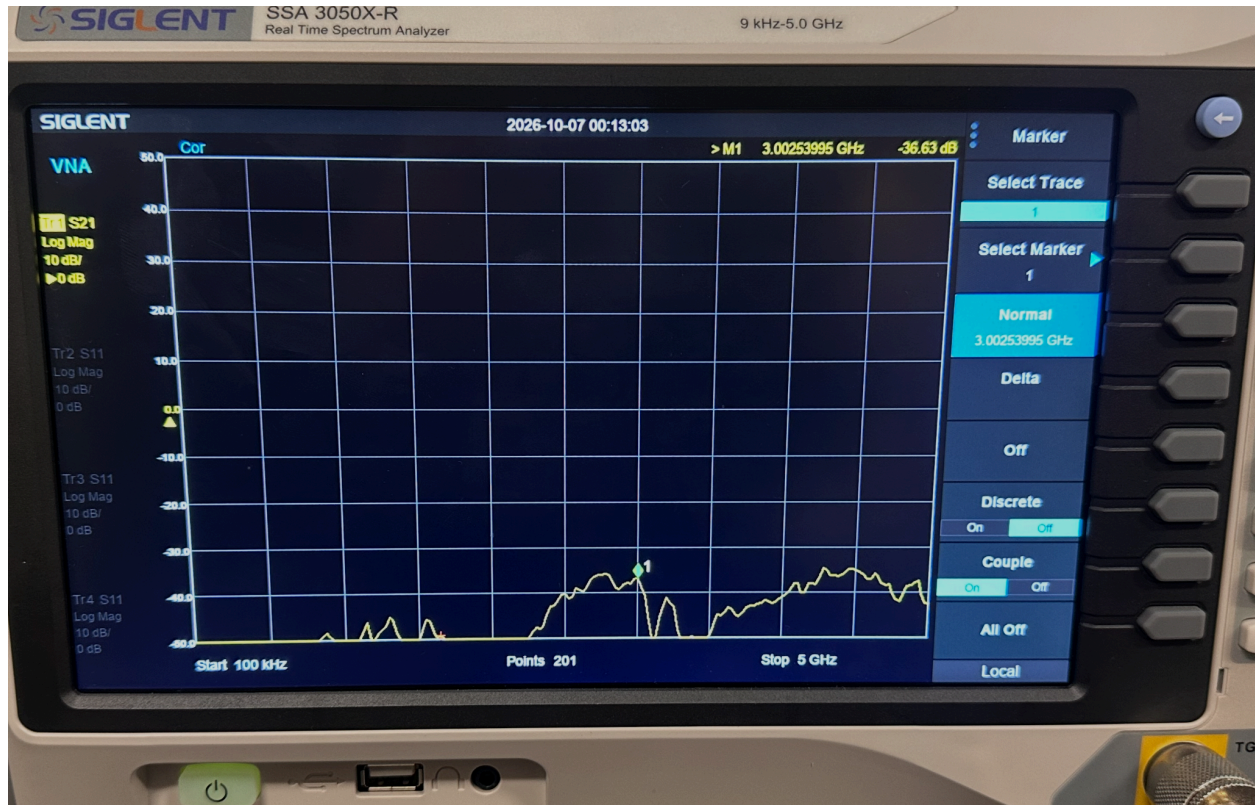


Figure 11: Measured S21 of the built filter, 100 kHz to 5 GHz.

The measured response does not look like a low-pass filter. From 100 kHz to roughly 2.3 GHz, where the simulation predicts S21 within 1.5 dB of 0 dB, the trace sits at the -50 dB bottom of the display. Transmission then rises to about -35/-30 dB around 2.5 - 3 GHz, and drops at around 3.2 GHz, and rises to a similar bump at around 4 GHz. The curve stays below -30 dB across the whole span. A low pass filter should pass low frequencies and attenuate the high ones, and here the highest frequencies get the most through.

Error Analysis:

The measured response did not match the simulation, and the sources of error fall into three groups. The first group can explain the complete loss of transmission. The second can only shift the response. The third comes from limitations in the design method and the measurement.

The most likely cause of the failure is a loss of electrical continuity somewhere in the signal path. The 0.41 mm inductor traces are very thin slivers of foil tape, and they can tear while being cut or peeled, or lift off the FR4. A single break in one of them would open the signal path. Each pad and trace was also a separate piece of tape joined to its neighbor by contact only, with no solder. A joint like this depends on direct metal-to-metal contact, and adhesive, oxide, or a tiny gap can make it highly resistive or fully open.

Other sources of error would have shifted the response without removing the passband. The traces were cut by hand, and it is difficult to cut foil tape to within a few tenths of a millimeter, especially for the 0.41 mm traces, where a small error is a large fraction of the width. The section lengths and pad widths also have some error, and the pieces may not be perfectly centered or aligned. These errors change the impedance and electrical length of each section, which moves the cutoff frequency and changes the stopband attenuation. The foil was also wrinkled, and its edges were uneven (see Figure 10). The adhesive layer under the tape adds a thin dielectric between the foil and the FR4 that is not in the model. Small gaps or overlaps at the joints add stray capacitance and inductance and change the effective section lengths. In addition, the permittivity and loss tangent of FR4 vary from board to board and with frequency, so the real board differs from the values used in the design.

Conclusion

In this lab, I used HFSS to match a complex load with a single open stub, match a $40\ \Omega$ load to a $75\ \Omega$ line with a quarter-wave transformer, and design a sixth-order stepped-impedance low-pass filter. The stub reduced the VSWR at 1 GHz from about 3.2 to 1.35, and the transformer reduced it at 2 GHz from about 1.34 to 1.17. Both matches worked only near the design frequency. The simulated filter had a cutoff of about 2.2 GHz, 12% below the 2.5 GHz target, and more than 20 dB of attenuation at 4 GHz. The filter I built did not show a low-pass response, and S21 stayed at or below -50 dB where the filter should have passed signal. This most likely came from a break or poor contact in the hand-cut foil tape, since cutting inaccuracy alone would only have shifted the cutoff. The lab showed that stubs, quarter-wave sections, and stepped-impedance filters work in simulation as they do in theory, and that hand-built microstrip needs far more precise construction to match.